5) اكتب برنامج لطباعة ~~[illegible]~~ المثلث الأعلى وبدلها برقم 1

$$\begin{bmatrix} 2 & 3 & 4 \\ 5 & 0 & 6 \\ 4 & 2 & 3 \end{bmatrix}$$

JavaApplication30 - NetBeans IDE 8.2

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JavaApplication29.java | JavaApplication30.java

Source History

```java
package javaapplication30;
import java.util.Scanner;
public class JavaApplication30 {
public static void main(String[] args) {
   // 2D Array
Scanner n=new Scanner(System.in);
int [][]a=new int[3][3];
int i,j;
System.out.println("Enter the number is :");
for(i=0;i<a.length;i++){
    for(j=0;j<a.length;j++)
        a[i][j]=n.nextInt();}
System.out.println("2D Array is :");
for(i=0;i<a.length;i++){
 for(j=0;j<a.length;j++)
     System.out.print(a[i][j]+" ");
    System.out.println();}
 System.out.println(" ------------------");
for(i=0;i<a.length;i++){
 for(j=0;j<a.length;j++)
     if(i+j<3-1)
     System.out.print(a[i][j]+" ");
    System.out.println();
}}}
```

Output - JavaApplication30 (run)

```
run:
Enter the number is :
2 3 4
5 0 6
4 2 3
2D Array is :
2 3 4
5 0 6
4 2 3
 ------------------
2 3
5

BUILD SUCCESSFUL (total time: 16 seconds)
```

Q2> Identify and correct the errors of the following set of code then print out the expected output: **(3 marks)**

```
String str = java language ;
System.out.println(str.lngth());
System.out.println(str.charat(4));
System.out.println(str.trim);
System.out.println(str.toUpperCase());
System.out.println(str ());
```

JavaApplication29.java

```java
package javaapplication29;
//import java.util.Scanner;
public class JavaApplication29 {
public static void main(String[] args)
{
  String str="java language"; // error ==> "            "
  System.out.println(str.length()); // error==> e
  System.out.println(str.charAt(4)); // error==> A
  System.out.println(str.trim()); // error==> ()
  System.out.println(str.toUpperCase()); // error==> not error
  System.out.println(str); // error==> ()
}
}
// output code
//--------------------
// 13
//
// java language
// JAVA LANGUAGE
// java language
```

Output - JavaApplication29 (run)

```
run:
13

java language
JAVA LANGUAGE
java language
BUILD SUCCESSFUL (total time: 0 seconds)
```

JavaApplication36 - NetBeans IDE 8.2

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JavaApplication33.java | JavaApplication34.java | JavaApplication35.java | JavaApplication36.java

Source History

```
int a[][]= new int [4][4];
Scanner x=new Scanner(System.in);
int i,j,t;
System.out.println("Enter the number:");
for(i=0;i<4;i++){
for(j=0;j<4;j++)
    a[i][j]=x.nextInt();
}
System.out.println("print arrary 2D ");
for(i=0;i<4;i++){
for(j=0;j<4;j++)
System.out.print(a[i][j]+" ");
System.out.println(" ");
}
for(i=0;i<4;i++){
for(j=0;j<4;j++)
if(i==j)
{
t=a[i][i];
a[i][i]=a[i][3-j];
a[i][3-j]=t;}}
System.out.println("--------------------- output 2D array: ");
for(i=0;i<4;i++){
for(j=0;j<4;j++)
System.out.print(a[i][j]+" ");
System.out.println(" ");

}
}
    }
```

Output - JavaApplication36 (run)

```
run:
Enter the number:
1 2 3 4
5 6 7 8
6 5 4 3
5 6 7 8
print arrary 2D
1 2 3 4
5 6 7 8
6 5 4 3
5 6 7 8
--------------------- output 2D array:
4 2 3 1
5 7 6 8
6 4 5 3
8 6 7 5
BUILD SUCCESSFUL (total time: 12 seconds)
```

31:59 INS

Group A: Write a suitable Java program to merge two arrays into a third.

For example: A= {2, 4}, B= {3, 6, 8} C= {2, 4, 3, 6, 8}

.....

Group C: Write a Java program to count prime numbers in an 1-D array.

B- Fill in the blanks in each of the following statements (**Answer 4 only**):

(4 marks)

1- The action of changing a data type into a different type is called _____.

2- The indexes of the array start with _____ always.

3- Method name _____ compares two strings. It returns true if these are identical, otherwise return false.

4- next() is a type of _____ methods.

5- The _____ statement terminates the current iteration of a loop and proceeds directly to the next instruction.

Q2 Fill in the blanks (5 marks)

1. The ---------- usually used with the switch-case statement; however, it can be used with a loop statement where it terminates the loop.
2. the string index begins at ----------.
3. length Returns the length of the string as ----------.
4. int a = (int)18.7; a=----------
5. A ---------- is an indexed list of values that all have the same type.

Q2> Fill in the blanks in each of the following statements: **(5 marks)**

1. The action of changing a data type into a different type is called _____.
2. The indexes of the array start with _____ always.
3. Method name _____ compares two strings. It returns true if these are identical, otherwise return false.
4. Array is a _____.
5. nextInt() is a type of _____ methods.

Q2 Fill in the blanks (5 marks)

1. In Java, the class --------- provides pre-written mathematical functions to perform various math calculations.
2. The method ---------- used to find the natural logarithm of x (base e).
3. System.out.println(Math.ceil(45.87)); output is----------
4. The method used to compare two strings without considering the case of small or capital letters is ----------
5. The ---------- statement skip the current iteration of a loop and proceeds directly to the next iteration.

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B- Fill in the blanks in each of the following statements (**Answer 4 only**):

(4 marks)

1- The action of changing a data type into a different type is called _____.

2- The indexes of the array start with _____ always.

3- Method name _____compares two strings. It returns true if these are identical, otherwise return false.

4- next() is a type of _____ methods.

5- The _____ statement terminates the current iteration of a loop and proceeds directly to the next instruction.

Q1> Determine whether each of the following is true or false: **(5 marks)**

1- If you create an array A= new int [3] you should think of it as a "*Array*" with 3 rows and 1 columns.
2- ceil(X) Rounds X to the largest integer not greater than X.
3- To access the elements of an array we use the { } operator.
4- Java is not an OOP (Object Oriented Programming) Language.
5- You can process all the elements of an array via a loop.

(5 marks)

Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextInt() method, which is used to read string. F
2. floor(x) method Rounds x to the smallest integer not less than x? F
3. To access the elements of an array we use the { } operator. T
4. equals method Removes any spaces at the beginning or end of a string. F
5. The output of charAt(i) method is int. F

5

Note: Attempt all questions

Q1> A- Determine whether each of the following is true or false (**Answer 4 only**): **(4 marks)**

1- If you create an array A= new int [5][6], you should think of it as a "*matrix*" with 6 rows and 4 columns.

2- exp(x): the natural logarithm of x.

3- To access the elements of an array we use the { } operator.

4- Java is not an OOP (Object Oriented Programming) Language.

5- You can process all the elements of an array via the switch-case statement.

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ment : Computer Science
Name : Advanced programming
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Academic Year : 2021 / 2022
Course Stage / Level : 1st
Date of Exam : 15 / 5 / 2022
Time : 1 Hours

Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextLine() method, which is used to read int.
2. ceil(x) method Rounds x to the largest integer not greater than x
1. To access the elements of an array we use the [] operator.
2. toUpperCase method Returns a copy of the original string, all lower case.
3. The output of Substring(i,j) method is int.

Q3> Identify and correct the errors of the following set of code (**Answer 2 only**):

(6 marks)

1)
```
int a=11.5; b=5.3;
System.out.println(math.Max(a, b));
System.out.println(math.abs(a));
```

2)
```
char ch = 'A';
int num = ch;
System.out.print(nume + cher);
```

3)
```
printSquare("hello");
printSquare(3.5)
public static void printSquare(int x) {
System.out.println(x*x);
}
```

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Department: Computer Science
Course Name: Advanced Programming

Course Stage/Level: 1st
Date of Exam: / /2021
Time: Two hours

Q2> Find the output of the following code (**Answer 2 only**): **(6 marks)**

1)
```
for (i = 1; i <= 9; i++ )
{
    for (j = 1; j <= i; j++ )
    {
        System.out.print((i+1)+" ");
    }
System.out.println();
}
```

2)
```
start = "Talk is cheap. ";
middle = "Show me the code. ";
end = "- Linus Torvalds";
result = start + middle + end;
System.out.println(result);
System.out.println(result.charAt(7));
```

3)
```
int[] Array = {2, 5, 1, 9, -11};
System.out.println(Array.length);
System.out.println(Array[-2]);
```

Q4> Identify and correct the errors of the following set of code: (6 marks)

1)
```
Scanner A= Scanner(System.out);
System.out.println("Enter the number:");
int n= Scanner.nextInt();
A.open();
```

2)
```
for(i<10; int = 1; i++)
System.in.println(n+"+"+i"="n+i);
```

Q1> A- Determine whether each of the following is true or false **(Answer 4 only):**

(4 marks)

1- One-dimensional array has two indices.

2- Method contains a set of instructions that perform well defined task and only runs when it called.

3- To access the elements of an array we use the [ ] operator.

4- Implicit type casting, a character variable can be truncated to an integer.

5- An array can store many different types of values.